STRONG APPROXIMATION PROPERTY FOR HENSELIAN G-RINGS

THE STACKS PROJECT AUTHORS

Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Popescu's theorem. theorem-popescu (carded L26819). Factors the subalgebra generated by the solution through a smooth R-algebra.
- Lifting a section along a smooth map. More on Algebra, lemma lift-section-smooth-morphism (cross-chapter, named). Produces the etale R -> R' with trivial residue extension and the map B -> R'.
- Henselian local rings admit sections of etale maps. Algebra, lemma characterize-henselian (cross-chapter, named). Retracts R' back to R, bringing the solution into R.

Source reference. Original reference 04GG: focused-seeds/geometry/S9-smo-theorem-approximation-property/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07M7: focused-seeds/geometry/S9-smo-theorem-approximation-property/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 07GC: sources/stacks-smoothing/pinned-original-source.tex, source lines 3330–3330. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1. *Let R be a Noetherian local ring. Let $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$. Suppose that $(a_1, \ldots, a_n) \in (R^\wedge)^n$ is a solution in $R^\wedge$. If R is a henselian G-ring, then for every integer N there exists a solution $(b_1, \ldots, b_n) \in R^n$ in R such that $a_i - b_i \in \mathfrak{m}^N R^\wedge$.*

Proof. Let $c_i \in R$ be an element such that $a_i - c_i \in \mathfrak{m}^N$. Choose generators $\mathfrak{m}^N = (d_1, \ldots, d_M)$. Write $a_i = c_i + \sum a_{i,l} d_l$. Consider the polynomial ring $R[x_{i,l}]$ and the elements

$$g_j = f_j(c_1 + \sum x_{1,l} d_l, \ldots, c_n + \sum x_{n,l} d_{n,l}) \in R[x_{i,l}]$$

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The system of equations $g_j = 0$ has the solution $(a_{i,l})$. Suppose that we can show that g_j as a solution $(b_{i,l})$ in R. Then it follows that $b_i = c_i + \sum b_{i,l} d_l$ is a solution of $f_j = 0$ which is congruent to a_i modulo $\mathfrak{m}^N$. Thus it suffices to show that solvability over $R^\wedge$ implies solvability over R.

Let $A \subset R^\wedge$ be the R-subalgebra generated by $a_1, \ldots, a_n$. Since we've assumed R is a G-ring, i.e., that $R \to R^\wedge$ is regular, we see that there exists a factorization

$$A \to B \to R^\wedge$$

with B smooth over R, see Theorem 07GC. Denote $\kappa = R/\mathfrak{m}$ the residue field. It is also the residue field of $R^\wedge$, so we get a commutative diagram

$$\begin{array}{ccc} B & \dashrightarrow & R' \\ \uparrow & \searrow & \downarrow \\ R & \longrightarrow & \kappa \end{array}$$

Since the vertical arrow is smooth, More on Algebra, Lemma 07M7 implies that there exists an étale ring map $R \to R'$ which induces an isomorphism $R/\mathfrak{m} \to R'/\mathfrak{m}R'$ and an R-algebra map $B \to R'$ making the diagram above commute. Since R is henselian we see that $R \to R'$ has a section, see Algebra, Lemma 04GG. Let $b_i \in R$ be the image of a_i under the ring maps $A \to B \to R' \to R$. Since all of these maps are R-algebra maps, we see that $(b_1, \ldots, b_n)$ is a solution in R. $\square$